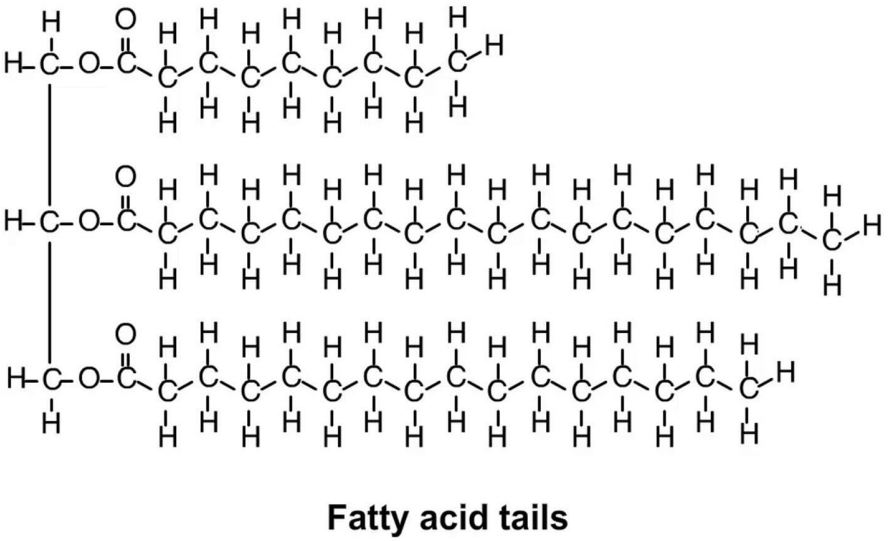
5.3: Lipids are a diverse group of hydrophobic molecules

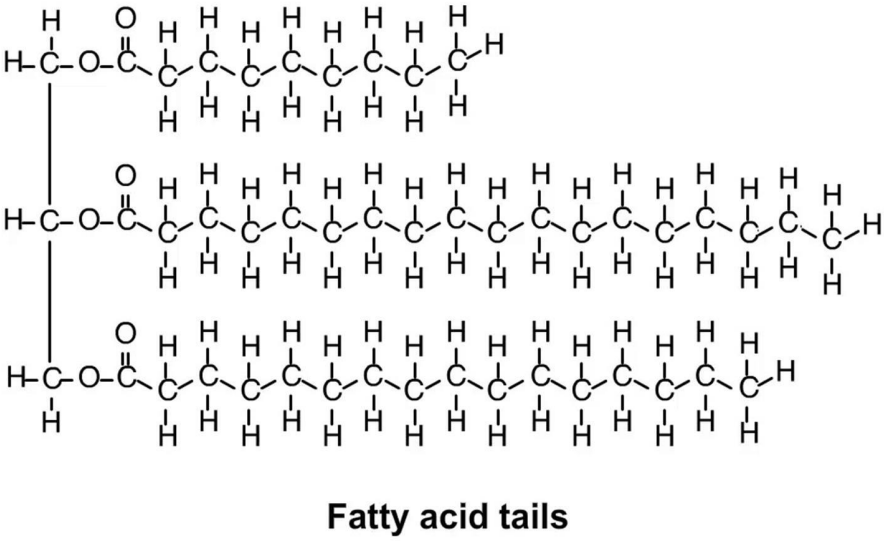
Lipids are the one class of large biological molecules that does not include true polymers, and they are generally not big enough to be considered macromolecules. The compounds called **lipids** are grouped with each other because they share one important trait: They are hydrophobic: They mix poorly, if at all, with water. This behavior of lipids is based on their molecular structure. Although they may have some polar bonds associated with oxygen, lipids consist mostly of hydrocarbon regions with relatively nonpolar C—H bonds. Lipids are varied in form and function. They include waxes and certain pigments, but we will focus on the types of lipids that are most important biologically: fats, phospholipids, and steroids.

Animation

Lipids



Play



Play

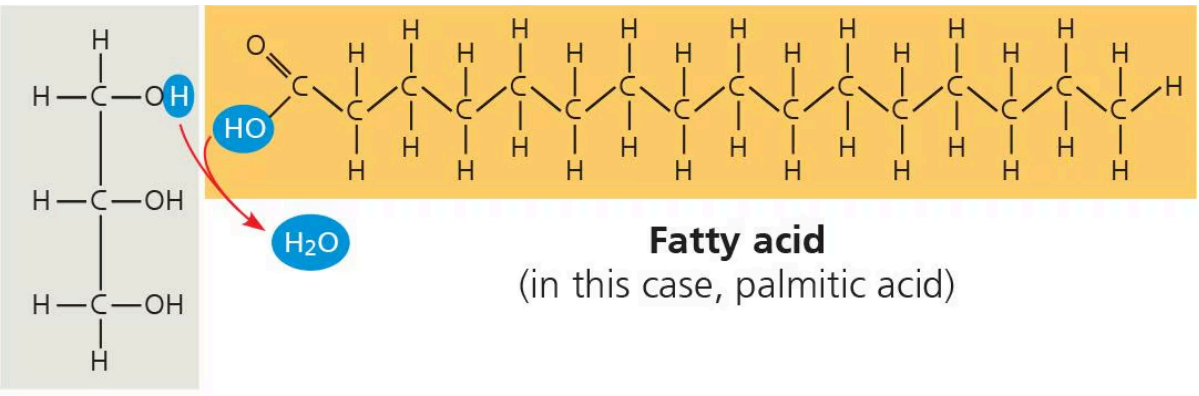
**

Fats

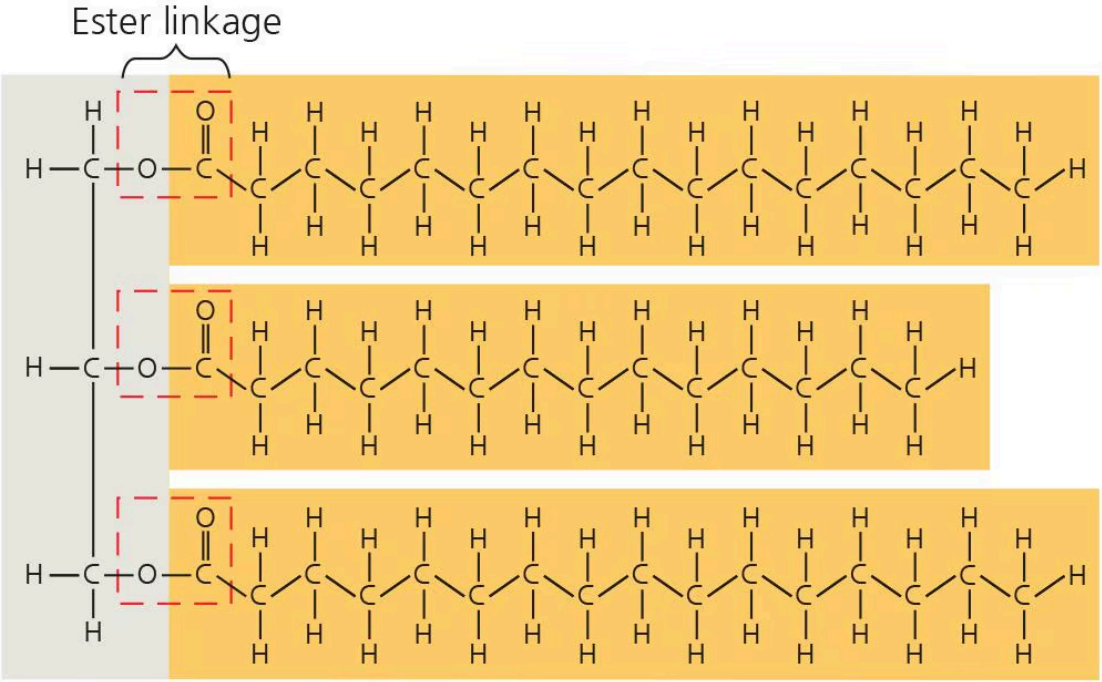
Although fats are not polymers, they are large molecules assembled from smaller molecules by dehydration reactions, like the dehydration reaction described in Figure 5.2a. A **fat** consists of a glycerol molecule joined to three fatty acids **(Figure 5.9)**. Glycerol is an alcohol; each of its three carbons bears a hydroxyl group. A **fatty acid** has a long carbon skeleton, usually 16 or 18 carbon atoms in length. The carbon at one end of the skeleton is part of a carboxyl group, the functional group that gives these molecules the name fatty *acid*. The rest of the skeleton consists of a hydrocarbon chain. The relatively nonpolar C—H bonds in the hydrocarbon chains of fatty acids are the reason fats are hydrophobic. Fats separate from water because the water molecules hydrogen-bond to one another and exclude the fats. This is why vegetable oil (a liquid fat) separates from the aqueous vinegar solution in a bottle of salad dressing.

Figure 5.9

The synthesis and structure of a fat, or triacylglycerol.



(a) One of three dehydration reactions in the synthesis of a fat. One water molecule is removed for each fatty acid joined to the glycerol.



(b) A fat molecule (triacylglycerol) with three fatty acid units. In this example, two of the fatty acid units are identical.

Part a. One of three dehydration reactions in the synthesis of a fat: synthesis of a triacylglycerol molecule. The formation of a triacylglycerol, or fat molecule, begins when a molecule of glycerol reacts with a fatty acid in a dehydration reaction. Glycerol is a three carbon alcohol with one O H group on each carbon. A fatty acid is a long carbon chain with a carboxyl (C O 2 H) group at one end. The O H of the glycerol loses its hydrogen

when it bonds to the carbon of the carboxyl group. The hydrogen from the glycerol and the OH from the carboxyl group leave and form a water molecule, and the glycerol and fatty acid are joined in an ester linkage.

Part b. Structure of a triacylglycerol (fat) molecule. A triacylglycerol is formed when a dehydration reaction occurs between each of the three OH groups on the glycerol molecule and a total of three fatty acid molecules. These fatty acids may be the same or different from one another. The resulting molecule of fat is held together by three ester linkages.

The molecular building blocks of a fat are one molecule of glycerol and three molecules of fatty acids. The carbons of the fatty acids are arranged zigzag to suggest the actual orientations of the four single bonds extending from each carbon (see Figures 4.3a and 4.6b).

In making a fat, each fatty acid molecule is joined to glycerol by a dehydration reaction **(Figure 5.9a)**. This results in an ester linkage, a bond between a hydroxyl group and a carboxyl group. The completed fat consists of three fatty acids linked to one glycerol molecule. (Other names for a fat are ***triacylglycerol*** and *triglyceride*; levels of triglycerides are reported when blood is tested for lipids.) The fatty acids in a fat can all be the same, or they can be of two or three different kinds, as in **Figure 5.9b**.

The terms *saturated* fats and *unsaturated* fats are commonly used in the context of nutrition **(Figure 5.10)**. These terms refer to the structure of the hydrocarbon chains of the fatty acids. If there are no double bonds between carbon atoms composing a chain, then as many hydrogen atoms as possible are bonded to the carbon skeleton. Such a structure is said to be *saturated* with hydrogen, and the resulting fatty acid is therefore called a **saturated fatty acid(Figure 5.10a)**. An **unsaturated fatty acid** has one or more double bonds, with one fewer hydrogen atom on each double-bonded carbon. Nearly every double bond in naturally occurring fatty acids is a *cis* double bond, which creates a kink in the hydrocarbon chain wherever it occurs **(Figure 5.10b)**. (See Figure 4.7b to remind yourself about *cis* and *trans* double bonds.)

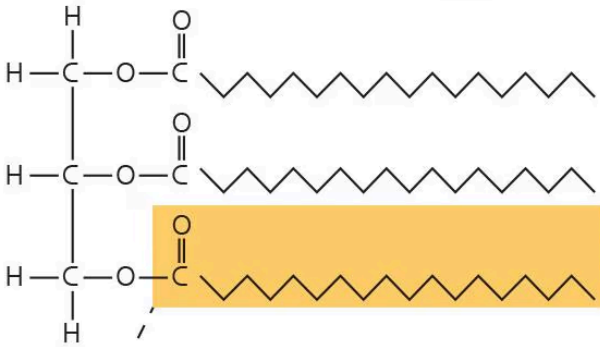
Figure 5.10

```
Saturated and unsaturated fats and fatty acids.
```

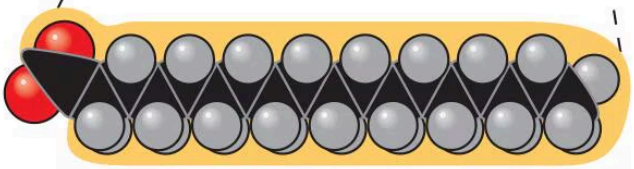
(a) Saturated fat

At room temperature, the molecules of a saturated fat, such as the fat in butter, are packed closely together, forming a solid.

Structural formula of a saturated fat molecule (Each hydrocarbon chain is represented as a zigzag line, where each bend represents a carbon atom; hydrogens are not shown.)



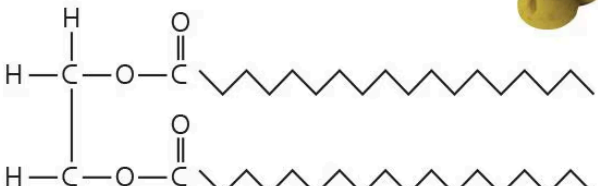
Space-filling model of stearic acid, a saturated fatty acid (red = oxygen, black = carbon, gray = hydrogen)

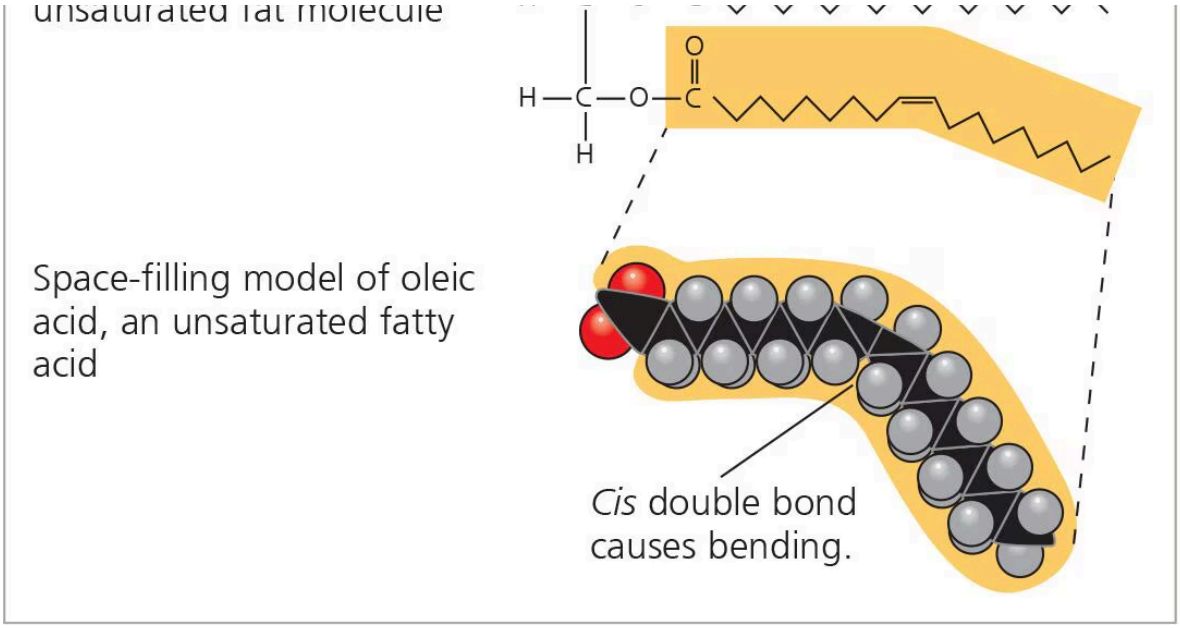


(b) Unsaturated fat

At room temperature, the molecules of an unsaturated fat such as olive oil cannot pack together closely enough to solidify because of the kinks in some of their fatty acid hydrocarbon chains.

Structural formula of an unsaturated fat molecule





Part a. At room temperature, the molecules of a saturated fat, such as the fat in butter, are packed closely together, forming a solid. In the structural formula of a saturated fat molecule, the fat is a glycerol joined to three fatty acid “tails” through ester linkages. Each hydrocarbon chain (tail) is represented as a zigzag line, where each bend represents a carbon atom and hydrogen atoms are not shown. The space-filling model of stearic acid, a saturated fatty acid, shows the carbon in the ester linkage bonded to one oxygen atom, double bonded to another oxygen atom, and then bonded to an unkinked (straight) chain of carbon and hydrogen atoms in the “tail” of the saturated fatty acid.

Part b. At room temperature, the molecules of an unsaturated fat such as olive oil cannot pack together closely enough to solidify because of the kinds in some of their fatty acid hydrocarbon chains. The structural formula of an unsaturated fat molecule (glycerol joined to three fatty acid tails) shows a double bond in one of the hydrocarbon chains. A space-filling model of oleic acid, an unsaturated fatty acid, shows that the double bond in the hydrocarbon chain depicted in the structural formula is a Cis double bond, which results in bending (a kink).

A fat made from saturated fatty acids is called a saturated fat. Most animal fats are saturated: The hydrocarbon chains of their fatty acids—the “tails” of the fat molecules—lack double bonds (see Figure 5.10a), and their flexibility allows the fat molecules to pack together tightly. Saturated animal fats, such as lard and butter, are solid at room temperature. In contrast, the fats of plants and fishes are generally unsaturated, meaning that they are composed of one or more types of unsaturated fatty acids. Usually liquid at room temperature, plant and fish fats are referred to as oils—olive oil and cod liver oil are examples. The kinks where the *cis* double bonds are located (see Figure 5.10b) prevent the molecules from packing together closely enough to solidify at room temperature. The phrase *hydrogenated vegetable oils* on food labels

means that unsaturated fats have been synthetically converted to saturated fats by adding hydrogen, allowing them to solidify. Peanut butter, margarine, and many other products are hydrogenated to prevent lipids from separating out in liquid (oil) form.

A diet rich in saturated fats is one of several factors that may contribute to the cardiovascular disease known as atherosclerosis. In this condition, deposits called plaques develop within the walls of blood vessels, causing inward bulges that impede blood flow and reduce the resilience of the vessels. The process of hydrogenating vegetable oils produces not only saturated fats but also unsaturated fats with *trans* double bonds. It appears that ***trans*fats** can contribute to coronary heart disease (see Concept 42.4). Because *trans* fats have been especially common in baked goods and processed foods, the U.S. Food and Drug Administration (FDA) requires nutritional labels to include information on *trans* fat content. In addition, the FDA has ordered U.S. food manufacturers to stop adding *trans* fats to foods. Globally, the World Health Organization has a stated goal of eliminating industrially produced *trans* fats by 2023; over a third of countries worldwide have already passed such regulations.

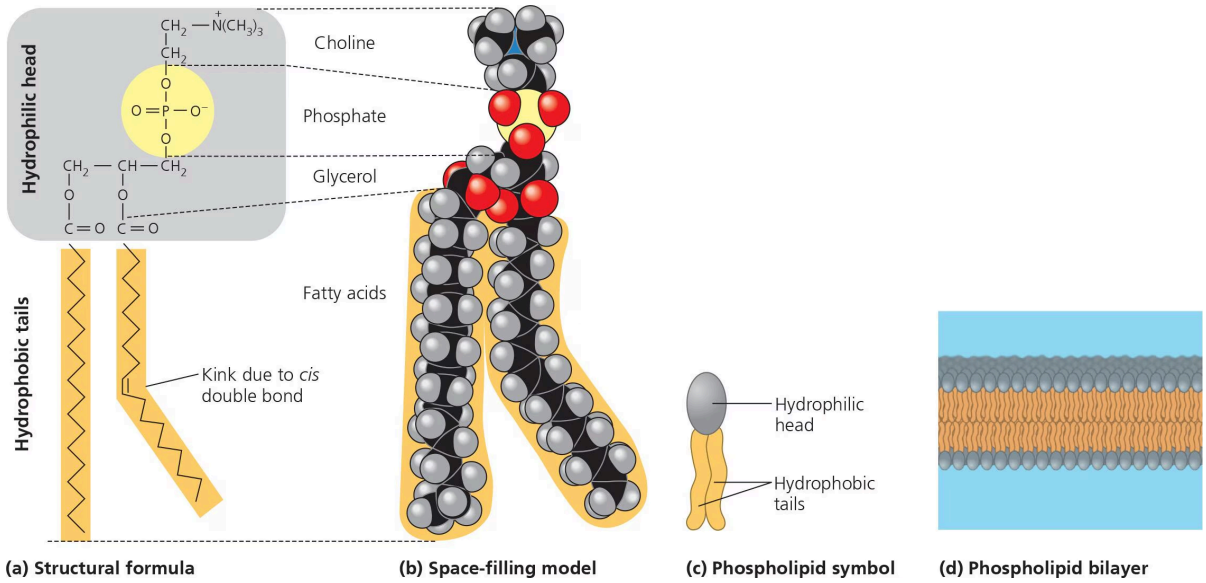
The major function of fats is energy storage. The hydrocarbon chains of fats are similar to gasoline molecules and just as rich in energy. A gram of fat stores more than twice as much energy as a gram of a polysaccharide, such as starch. Because plants are relatively immobile, they can function with bulky energy storage in the form of starch. (Vegetable oils are generally obtained from seeds, where more compact storage is an asset to the plant.) Animals, however, must carry their energy stores with them, so there is an advantage to having a more compact reservoir of fuel—fat. Humans and other mammals stock their long-term food reserves in adipose cells (see Figure 4.6a), which swell and shrink as fat is deposited and withdrawn from storage. In addition to storing energy, adipose tissue also cushions such vital organs as the kidneys, and a layer of fat beneath the skin insulates the body. This subcutaneous layer is especially thick in whales, seals, and most other marine mammals, insulating their bodies in cold ocean water.

Phospholipids

Cells as we know them could not exist without another type of lipid, called phospholipids. Phospholipids are essential for cells because they are major constituents of cell membranes. Their structure provides a classic example of how form fits function at the molecular level. As shown in **Figure 5.11**, a **phospholipid** is similar to a fat molecule but has only two fatty acids attached to glycerol rather than three. The third hydroxyl group of glycerol is joined to a phosphate group, which has a negative electrical charge in the cell. Typically, an additional small charged or polar molecule is also linked to the phosphate group. Choline is one such molecule (see Figure 5.11), but there are many others as well, allowing formation of a variety of phospholipids that differ from each other.

Figure 5.11

The structure of a phospholipid.



Part a. Structural formula of a phospholipid. A charged molecule is linked to a negatively charged phosphate group, which is linked to a glycerol attached to two hydrophobic fatty acid tails. The phospholipid consists of a hydrophilic (charged) head and two hydrophobic tails. Choline (a nitrogen-containing molecule with a positive charge on N) is linked to a phosphate group (a central P double bonded to one O and single bonded to three other oxygen atoms with a negative charge on one oxygen atom). The phosphate group is joined to the third hydroxyl group of glycerol, which is attached to two fatty acid tails (hydrophobic chains of carbon and hydrogen; in one tail there is a kink due to a cis double bond).

Part b. Space filling model of a phospholipid. Choline molecule contains N and is linked to the phosphate group containing P, which is linked to a glycerol attached to two fatty acids.

Part c. Phospholipid symbol: An oval represents the hydrophilic head; is attached to two squiggly lines representing the hydrophobic tails.

Part d. Phospholipid bilayer drawn with phospholipid symbols. Heads contact water on both sides of the membrane, tails point at each other inside the membrane.

A phospholipid has a hydrophilic (polar) head and two hydrophobic (nonpolar) tails. This particular phospholipid, called a phosphatidylcholine, has a choline attached to a phosphate group. Shown here are **(a)** the structural formula, **(b)** the space-filling model (yellow = phosphorus, blue = nitrogen), **(c)** the symbol for a phospholipid that will appear throughout this book, and **(d)** the bilayer structure formed by selfassembly of phospholipids in an aqueous environment.

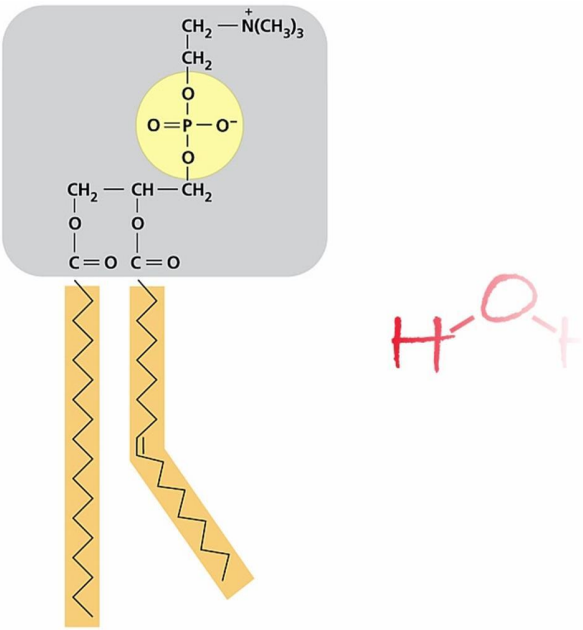
DRAW IT *Draw an oval around the hydrophilic head of the space-filling model.*

*For suggested answer, see**Appendix A*.

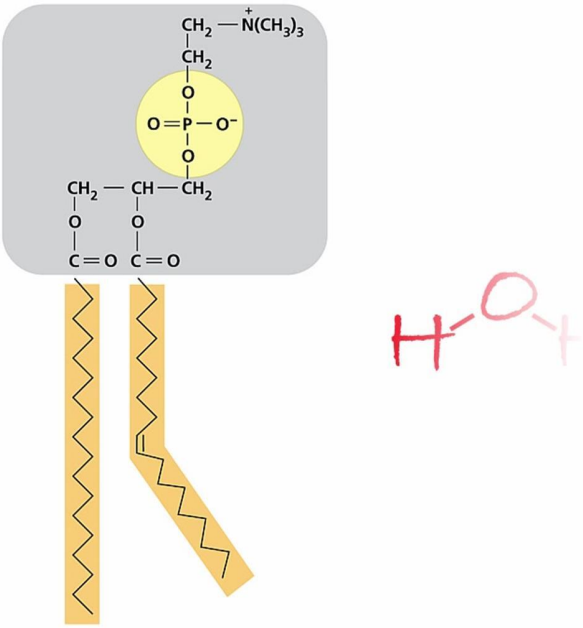
Figure 5-11 Walkthrough

The structure of a phospholipid

Watch an explanation of this figure in the Figure 5.11 Walkthrough: The structure of a phospholipid.



Play

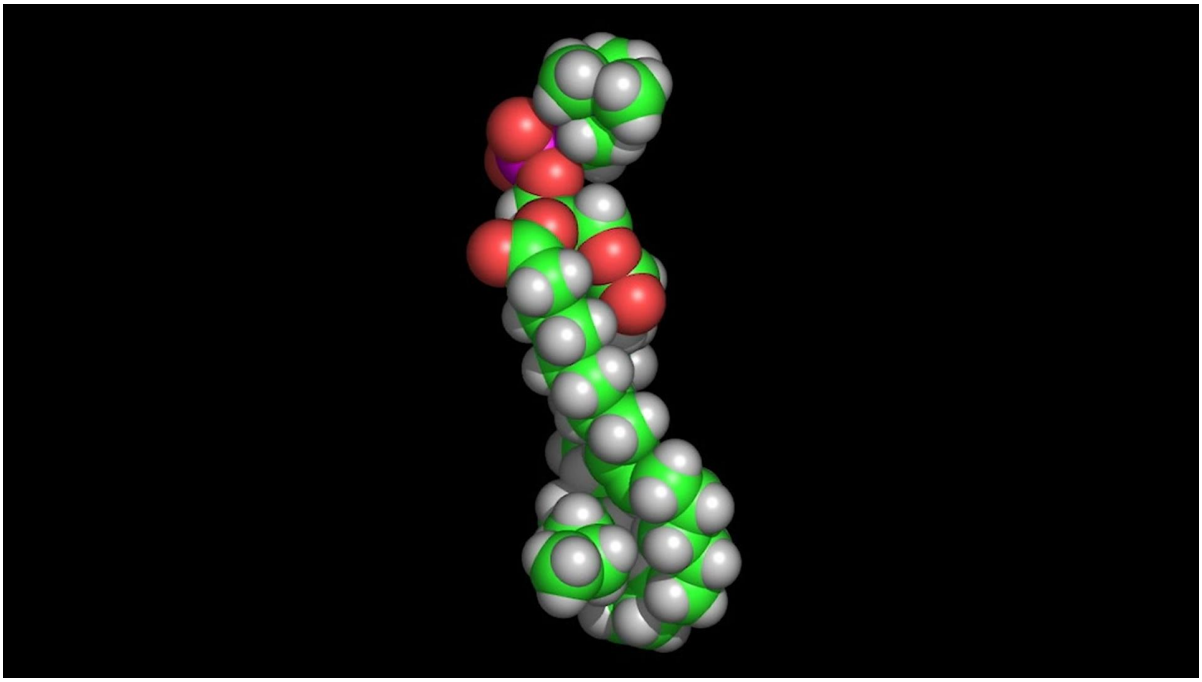


Play

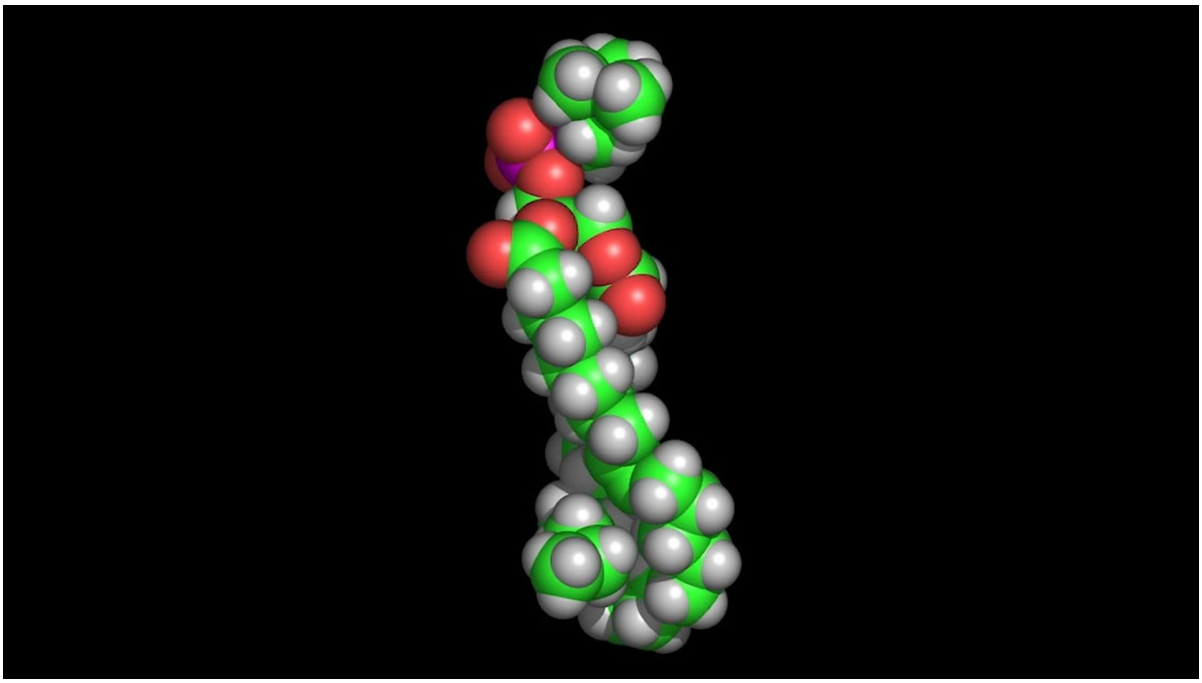
**

Animation

Space-Filling Model of a Phospholipid



Play



Play

The two ends of phospholipids show different behaviors with respect to water. The hydrocarbon tails are hydrophobic and are excluded from water. However, the phosphate group and its attachments form a hydrophilic head that has an affinity for water. When phospholipids are added to water, they self-assemble into a double-layered sheet called a “bilayer” that shields their hydrophobic fatty acid tails from water (Figure 5.11d).

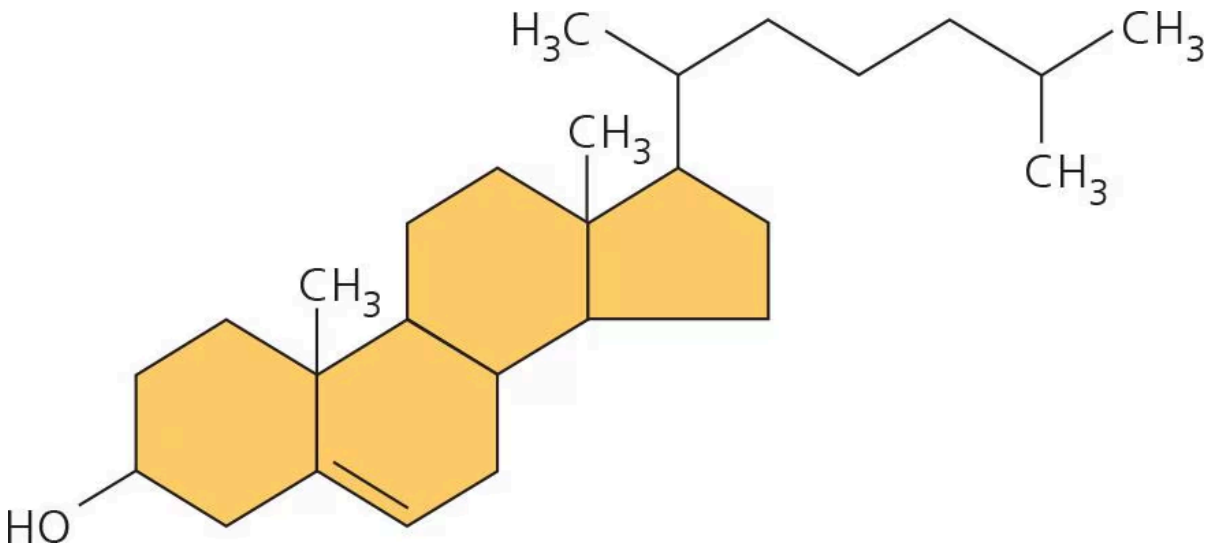
At the surface of a cell, phospholipids are arranged in a similar bilayer. The hydrophilic heads of the molecules are on the outside of the bilayer, in contact with the aqueous solutions inside and outside of the cell. The hydrophobic tails point toward the interior of the bilayer, away from the water. The phospholipid bilayer forms a boundary between the cell and its external environment and establishes separate compartments within eukaryotic cells; in fact, the existence of cells depends on the properties of phospholipids.

Steroids

Steroids are lipids characterized by a carbon skeleton consisting of four fused rings. Different steroids are distinguished by the particular chemical groups attached to this ensemble of rings. **Cholesterol**, a type of steroid, is a crucial molecule in animals **(Figure 5.12)**. It is a common component of animal cell membranes and is also the precursor from which other steroids, such as the vertebrate sex hormones, are synthesized. In vertebrates, cholesterol is synthesized in the liver and is also obtained from the diet. A high level of cholesterol in the blood may contribute to atherosclerosis, although some researchers are questioning the roles of cholesterol and saturated fats in the development of this condition.

Figure 5.12

Cholesterol, a steroid.



Cholesterol is the molecule from which other steroids, including the sex hormones, are synthesized. Steroids vary in the chemical groups attached to their four interconnected rings (shown in gold).

Cholesterol contains the four fused rings of a steroid. It has an alcohol on the A ring, a branched 8-carbon tail on the D ring, and a methyl group at both the AB and CD ring junctions. There is also one double bond in the B ring. All other steroids are made from cholesterol.

MAKE CONNECTIONS *Compare cholesterol with the sex hormones shown in the figure at the beginning of*Concept 4.3*. Circle the chemical groups that cholesterol has in common with estradiol; put a square around the chemical groups that cholesterol has in common with testosterone.*

*For suggested answer, see**Appendix A*.

Interview

Interview with Lovell Jones: Investigating the effects of sex hormones on cancer